



Cooperation dynamics on hypergraphs with punishment and Q-learning

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ABSTRACT

Punishment has been proved to be a useful mechanism on pairwise interaction networks for promoting cooperation. However, these networks cannot effectively describe higher-order interactions in the multi-agent system, while hypergraph as a higher-order network has aroused extensive interests of researchers. Here, we study the evolutionary dynamics of spatial public goods game on uniform random hypergraphs with peer punishment mechanism. In that model, each agent chooses to become a cooperator, defector or punisher, and each punisher pay a cost to make each defector bear a fine. Different from the imitation rules in previous studies, we adopt self-regarding Q-learning algorithm to update agent's strategy where agents take an action based on their historical experience to maximize their reward. Simulation results show that there is a moderate synergy factor can obtain the best result of the evolution of cooperation. For a certain relatively large synergy factor, there exists a combination of cost and fine to optimally promote cooperation. Furthermore, the theoretical analysis results are consistent with the simulation results.

1. Introduction

Cooperation is prevalent in biological and social systems (Axelrod & Hamilton, 1981). In the real society, altruistic cooperation is an action in which an agent provides a benefit to another agent by paying a cost. However, many rational agents will take selfish behavior to maximize their own benefits because of the natural selection of Darwinian theory, which will cause a cooperation dilemma for the whole population (Perc et al., 2017; Rand & Nowak, 2013). So far, evolutionary game theory (Allen et al., 2017) has been applied to many social dilemmas, and several typical models have been proposed to study the emergence and maintenance of cooperation, such as prisoner's dilemma (PDG) (Nowak & May, 1992; Wang et al., 2018, 2017), snowdrift game (SDG) (Hauert & Doebeli, 2004; Santos et al., 2012; Zhang & Huang, 2023), public goods game (PGG) (Hua & Liu, 2024; Santos et al., 2008; Yang & Yang, 2019), and other games (Han & Li, 2022; Ma et al., 2023).

As a classical model for understanding group interactions in multi-agent systems, spatial public goods game (SPGG) assumes that each agent chooses either to cooperate with other agents by contributing a fixed investment to the common pool or to defect by paying nothing (Szabó & Fáth, 2007). Then the total investment is increased by multiplying a synergy factor, and all agents share this amount evenly whatever their strategies. In this scenario, defection is the best action to obtain higher benefits from cooperators (Wang et al., 2020). Consequently, a series

of cooperation mechanisms have been proposed to uncover the evolution of cooperation in the social dilemma, for example, reward (Szolnoki & Perc, 2010; Wang et al., 2014), reputation (Fu et al., 2008; Xia et al., 2023), and exclusion (Liu et al., 2017, 2018). Besides, punishment as a mechanism is introduced to explain the emergence of cooperation (Wang et al., 2023b), and it has been proved to be an important method for promoting cooperative behavior in human societies by punishing defectors when punishers are present in groups. Indeed, there are agents who advocate for fairness and are willing to punish those who seek excessive benefits through unfair methods (Henrich et al., 2006; Sigmund, 2007). In addition to the efforts of these agents, several sanctioning institutions have been established in reality to penalize unfair behaviors, such as tax evasion (Güerker et al., 2006). These two cases of punishment are referred to as peer punishment (Flores et al., 2021) and pool punishment (Szolnoki et al., 2011), respectively. Traditionally, peer punitive behavior is considered to be altruistic and costly, namely, a punisher not only contributes to the investment like a cooperator but also incurs a cost to punish a defector who then bears a fine. Helbing et al. (2010) considered the effects of punishing cooperators versus defectors as an additional strategy to study the evolution of cooperation in SPGGs, and found that the cooperation level shows a non-monotonous dependence on the fine. Chen et al. (2014) demonstrated that distributing the responsibility for sanctioning defectors, rather than relying solely on specific individuals, can effectively address the issue of costly punishment.

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Quan et al. (2023) introduced a novel graded punishment within the SPGG and demonstrated that graded punishment generally fosters greater cooperation than conventional probabilistic punishment. Nakamaru and Yokoyama (2020) examined how ostracism (functioning as costless punishment) versus group participation mechanisms influence cooperation dynamics in voluntary public goods games. They demonstrated that the maintenance of cooperative behavior depends critically on ostracism's ability to sanction defectors without incurring costs, rather than on simple participation in group interactions.

Up to now, most previous studies on punishment mechanisms have adopted imitation rules to update an agent's strategy by comparing the benefits between the agent and its neighbor. But these imitation rules disregard an agent's personal experience from multiple rounds of games. In recent years, reinforcement learning (Horita et al., 2017; Pendharkar, 2012) has been an effective method to renew agents' strategies based on their personal experience. As the two classical methods of reinforcement learning, the Bush-Mostelle (BM) model (Macy & Flache, 2002) and the Q-learning algorithm (Watkins & Dayan, 1992) have gradually been used to establish a reasonable framework to study the dynamical process of the evolutionary game. Jia et al. (2021) investigated pattern formation and phase transitions toward cooperation in social dilemmas influenced by the BM model. Their findings highlight the crucial role of global players in promoting cooperation. Song et al. (2022) integrated the iterated PDG with the BM model and demonstrated that moderate switching dynamics in interaction intensity foster optimal evolution of cooperation. Shi and Rong (2022) explored the dynamics of Q-learning and Frequency Adjusted Q-learning (FAQ-learning) algorithms in multi-agent systems and observed that the FAQ-learning algorithm can converge to the equilibrium state of a game model. Wang et al. (2023a) integrated the PGG with self-regarding Q-learning and adaptive reward mechanism, revealing a substantial increase in the cooperation level when employing the adaptive reward strategy.

However, the aforementioned studies assume that agents within a group are only pairwise connected to a central agent, and all other interactions between agents within the same group occur solely through this central agent (Battiston et al., 2020). Moreover, in human societies, interactions are not only one-to-one but also one-to-many (higher-order interactions), which cannot be well explained by simple topologies. These phenomena indicate that classical networks are not suitable for addressing higher-order interactions (Battiston et al., 2021; Majhi et al., 2022). As one of the typical higher-order networks, hypergraph can effectively encode group interactions of three or more agents, which are suited for modeling this complex interaction. Recently, many studies have focused on the evolution of agents' behavior on hypergraphs for various social dilemmas, such as the sender-receiver game (Kumar et al., 2021), group choice dilemma (Civilini et al., 2021), metanorms game (Wang et al., 2022), naming game (Iacopini et al., 2022), public goods game (Alvarez-Rodriguez et al., 2021; Burgio et al., 2020; Pan et al., 2023; Zou et al., 2024; Zou & Huang, 2024), and others.

Existing studies on punishment mechanisms in the evolution of cooperation demonstrate that introducing punitive behavior can generate optimal cooperation phenomena at specific cost-fine ratios, and concurrent studies on self-regarding Q-learning algorithm in cooperative systems reveal that this algorithm effectively captures how distinct mechanisms produce optimized cooperative outcomes under varying conditions. We are interested in the evolutionary dynamics of cooperation when the punishment mechanism and self-regarding Q-learning algorithm are considered at the same time. We assume that the public goods game model incorporating both punishment mechanism and self-regarding Q-learning algorithm will similarly exhibit optimal cooperation phenomena on hypergraphs. Inspired by the aforementioned works, in this study, we propose a peer punishment-based spatial public goods game on uniform random hypergraphs, where three strategies of agent (i.e., cooperation, defection and punishment) are updated by self-regarding Q-learning algorithm. In this model, each agent aims to maximize their reward based on accumulated experience.

We assume that each punisher incurs a cost to make each defector bear a fine, thus the total cost (fine) for a punisher (defector) is determined by the number of defectors (punishers) in the group. When defectors are absence in a group, the benefit for punishers is equivalent to that of cooperators. Simulation results show that a moderate synergy factor can achieve the highest cooperation level, regardless of the values of cost and fine. Besides, the effect of cost or fine is contingent upon the value of another parameter, and there is an optimal combination of cost and fine that promotes the evolution of cooperation for a given synergy factor. Moreover, a moderate cost can keep the frequency of punishers relatively low to punish defector, and a moderate fine can maximize the fraction of fully cooperative clusters. Furthermore, we find that randomness in decision-making can significantly affect the evolution of cooperation. Finally, we provide a theoretical analysis of the model, and the theoretical analysis results are consistent with the simulation results.

The rest of this paper is organized as follows. Section 2 introduces our model, which includes the evolutionary spatial public goods game with punishers on uniform random hypergraphs and the self-regarding Q-learning algorithm. Section 3 provides a comprehensive analysis of numerical simulation results, including the effects of different values of cost and fine. Finally, Section 4 summarizes the paper and provides a conclusion.

2. Model

In our model, the uniform random hypergraph (URH), as a higher-order network, is constructed with N nodes and L hyperlinks to describe the intricate relationships among all agents in a group. Unlike classical networks, a hyperlink in the URH contains two or more nodes and each hyperlink has the same number of nodes, referred to as the order g of the URH. Each node represents an agent randomly assigned to a hyperlink, and the number of hyperlinks in which an agent i participates is denoted as its hyperdegree k_i . Consequently, each agent i belongs to k_i overlapping groups containing g agents, and the average hyperdegree of the whole population can be calculated as $\langle k \rangle = \frac{1}{N} \sum_{i \in N} k_i$. To ensure a connected URH (Alvarez-Rodriguez et al., 2021), the number of hyperlinks and average hyperdegree should satisfy the critical values as $L \geq L_c = (N/g) \ln N$ and $\langle k \rangle \geq k_c = \ln N$, respectively. Ghanetto and Heydari (2016) revealed that sparse clique structures exert a more pronounced influence on the evolutionary dynamics of cooperative behavior, and our model demonstrates robust performance across varying network scales, exhibiting consistent behavioral patterns regardless of system size. Thus, we set up the network parameters of this model as $N = 1000$, $g = 5$, $L = L_c$, and $\langle k \rangle = k_c$.

In the spatial public goods game (SPGG) involving peer punishment, each cooperator (C) or punisher (P) in a hyperlink contributes an investment c to the common pool, while defectors (D) contribute nothing. The summation of contributions is multiplied by a synergy factor R and then evenly distributed among all agents in the hyperlink. Subsequently, if the hyperlink includes both a punisher and a defector, each punisher is required to incur a cost α and each defector is punished by a fine β . Therefore, the payoffs of different strategies in the hyperlink h are depicted as

$$\begin{aligned}\pi_C^h &= r(N_C + N_P)c - c, \\ \pi_D^h &= r(N_C + N_P)c - \beta N_P, \\ \pi_P^h &= r(N_C + N_P)c - c - \alpha N_D,\end{aligned}\quad (1)$$

where $r = R/g$ is the normalized synergy factor and g is the hypergraph order. N_C , N_D and N_P are the number of cooperators, defectors and punishers in the hyperlink h , respectively. In general, the value of the investment c is fixed as 1. In terms of these payoffs, cooperative behavior can be viewed as a form of second-order free-riding behavior by exploiting punishers, and the payoffs of punishers are equal to those of cooperators only when there are no defectors present in the hyperlink. To eliminate the influence of varying hyperdegrees, the total payoffs

from all hyperlinks in which agent i is involved are averaged by i 's hyperdegree to obtain the normalized payoff Π_i , which can be expressed as

$$\Pi_i = \frac{1}{k_i} \sum_{j=1}^{k_i} \pi_i^j. \quad (2)$$

In our model, we use self-regarding Q-learning algorithm to update agent's strategy. Each agent has a state that reflects its personal experience from the previous round of the SPGG. Specifically, each state of an agent corresponds to a value for each action, known as the Q-value. All Q-values for various state-action pairs are organized in a Q-table, where rows denote the state set and columns denote the action set. We assume the state and action sets are $S = A = \{C, D, P\}$ and the Q-table of agent i is provided as

$$Q_{s,a}^i = \begin{bmatrix} Q_{C,C}^i & Q_{C,D}^i & Q_{C,P}^i \\ Q_{D,C}^i & Q_{D,D}^i & Q_{D,P}^i \\ Q_{P,C}^i & Q_{P,D}^i & Q_{P,P}^i \end{bmatrix}, \quad (3)$$

where $Q_{s,a}^i$ represents the Q-value for agent i 's in the combination of state s and action a . After obtaining the normalized payoff through the selected action, this payoff serves as the reward derived from environment and the current action a is the next state s' . The Q-table of agent i will be updated as

$$Q_{s,a}^i(t+1) = (1-\eta)Q_{s,a}^i(t) + \eta \left[\Pi_i + \gamma \max_{a'} Q_{s',a'}^i(t) \right], \quad (4)$$

where $\eta \in (0, 1]$ is the learning rate that reflects the emphasis on accumulated historical information, $\gamma \in [0, 1)$ is the discount factor that controls the impact of expected future reward on subsequent decision-making, and $\max_{a'} Q_{s',a'}^i$ is the highest Q-value among all potential actions in the next state s' .

We study the evolutionary dynamics of cooperation by using Monte Carlo simulations, including the following micro-steps. Initially, each agent is randomly assigned a state from the state set, and the Q-tables of all agents are initialized to 0. At each Monte Carlo step (MCS), an agent is randomly selected to perform action selection, payoff computation, and state transition, and these above procedures are repeated for N rounds. For action selection, each agent follows the ϵ -greedy algorithm to choose an action. Specifically, agents choose the action with the highest Q-value in current state with a probability of $1 - \epsilon$, or randomly choose an action from the action set with a probability of ϵ . Then, the selected agent plays spatial public goods game with all its neighbors to calculate game payoff. Finally, this agent updates current Q-value based on Eq. (4) according to agent's current state and action and transits state from s to s' . Algorithm 1 outlines all the detailed Monte Carlo simulation in our model. The system reaches a steady state by enduring 5×10^5 MCSs, and the stationary frequencies of different strategies in the whole population (f_C , f_D and f_P) are averaged over the last 5000 steps. Finally, we conduct 20 independent runs to obtain the final results.

3. Simulation results and analysis

Cost α and fine β are two key parameters of our structured model that play different roles in the evolution of cooperation under varying dilemma intensities r . To examine the effect of these parameters on cooperation, we first present three phase diagrams: $\alpha - r$, $\beta - r$ and $\beta - \alpha$, each depicting the frequency of the three strategies. As shown in Fig. 1(a-c), there is a moderate r that can lead to the highest cooperation level at a certain α when $\alpha > 0.58$. Additionally, when the dilemma intensity is larger ($r < 0.3$), cooperators cannot survive in the population, regardless of the value of α . As r increases from 0.3 to 1.16, punishers begin to punish defectors and there exists an optimal value of α that achieves the highest frequency of cooperators at a given r . These phenomena mean that cooperators expand their survival area with lower dilemma intensity and moderate punishment cost, while excessive cost

Algorithm 1 Evolutionary cooperation dynamics with punishment and reinforcement learning.

Input: Agent set N , State set S , Action set A , Total Monte Carlo step T .

Output: Frequency of different strategies, f_C , f_D and f_P .

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1: Build a URH;
2: Initialize Q-tables of all agents to 0;
3: for each agent  $i \in N$  do
4:   Randomly initialize the state  $s$  of  $i$  from  $S$ ;
5: end for
6: for each Monte Carlo step  $t \in [1, T]$  do
7:   for each round  $\in [1, N]$  do
8:     Randomly select an agent  $i$ ;
9:     Generate a random number prob between 0 and 1;
10:    if prob <  $\epsilon$  then
11:      Randomly choose an action  $a$  from  $A$ ;
12:    else
13:      Choose the action  $a$  with the highest Q-value in current state  $s$ ;
14:    end if
15:    Calculate  $i$ 's payoff and normalized payoff according to Eq. (1) and Eq. (2), respectively;
16:    Update  $i$ 's Q-table according to Eq. (4);
17:    Update  $i$ 's next state  $s'$  to current action  $a$ ;
18:    Transfer  $i$ 's state from  $s$  to  $s'$ ;
19:  end for
20: end for

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can lead to double exploitation of punishers by both cooperators and defectors. However, when $r > 1.16$, increasing α leads to the best result of cooperation, which suggests that punishers can benefit sufficiently to continue punishing defectors when r is large enough, and higher costs will reduce the proportion of punishers, making cooperation the dominant strategy. From the result of $\beta - r$ plane depicted in Fig. 1(d-f), we find that for a specific value of β , there exists an optimal r that maximizes cooperation. Besides, when $0.42 < r < 1.06$, cooperators emerge in the system if β is large enough and increasing β can promote the evolution of cooperation because defectors bear a large fine. Moreover, the critical value of β needed to maintain cooperative behavior decreases as r increases, but a small fine may allow defectors to evade the control of punishers. After punishers obtain substantial benefits from a large r , a small fine leads to the competition between defectors and punishers, which most effectively fosters cooperation, much like the saying "The wolf has a winning game when the shepherds quarrel". Furthermore, we provide $\beta - \alpha$ phase diagrams to examine the relationship between cost α and fine β . As shown in Fig. 1(g-i), there is a combination of α and β that achieves the best result for cooperation when α is moderate and β is small. As β increases, a larger fine results in a higher frequency of punishers and the disappearance of defectors, while increasing α can yield the better evolution of defection and leads to the disappearance of punishers. The critical thresholds of β at which defectors cease to exist and punishers emerge both increase with an increase in α . As demonstrated in the study of Benichmol and Palumbo (2024), the emergence of punishment and sanction behaviors can induce significant fluctuations and trend variations in collective cooperation level. Based on these observations, we summarize that there exist an optimal value of r can obtain the highest cooperation level, and there exist a certain value of $\alpha - \beta$ pair can be most conducive to the evolution of cooperation.

To intuitively observe these optimal cooperation results, we depict the fraction of cooperators as a function of the synergy factor r , cost α , and fine β in Fig. 2. From Fig. 2(a), we can observe that across various $\alpha - \beta$ pairs, there exists an optimal value of r that most effectively fosters cooperation, irrespective of the specific values of the $\alpha - \beta$ pair. When α is set to 0.5, the optimal value of r decreases with an increase in β . Conversely, when β is fixed at 0.5, the optimal value of r increases

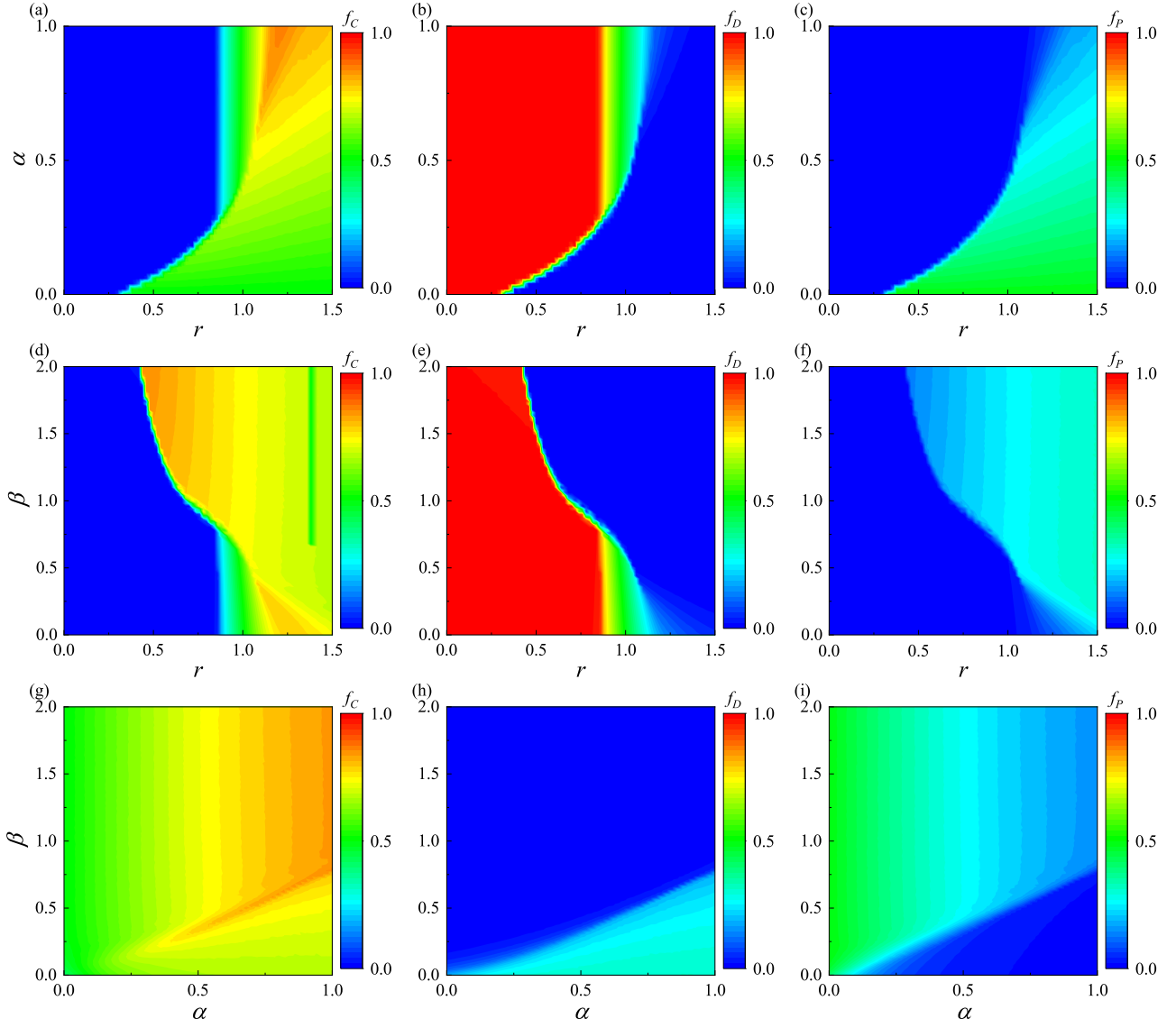


Fig. 1. Phase diagrams of $\alpha - r$, $\beta - r$, and $\beta - \alpha$ for the evolution of three strategies. (a–c), (d–f) and (g–i) represent the results of f_C , f_D and f_P under different combinations (i.e., $\alpha - r$ in $\beta = 0.5$, $\beta - r$ in $\alpha = 0.5$, and $\beta - \alpha$ in $r = 1.1$), respectively. Other parameters are $g = 5$, $L = L_c$, $\langle k \rangle = k_c$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

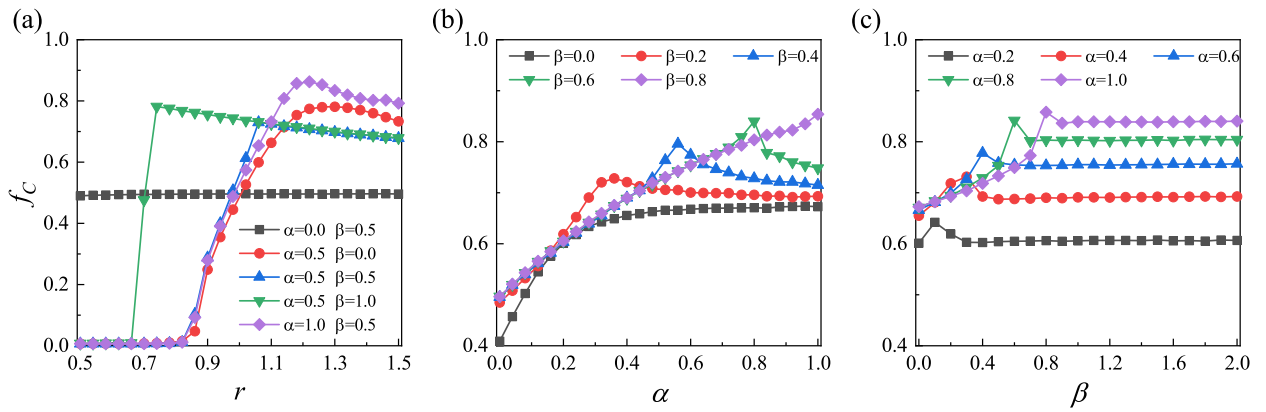


Fig. 2. Frequency of cooperators f_C as a function of r , α , and β for several values of another parameter. Other parameters are $g = 5$, $L = L_c$, $\langle k \rangle = k_c$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

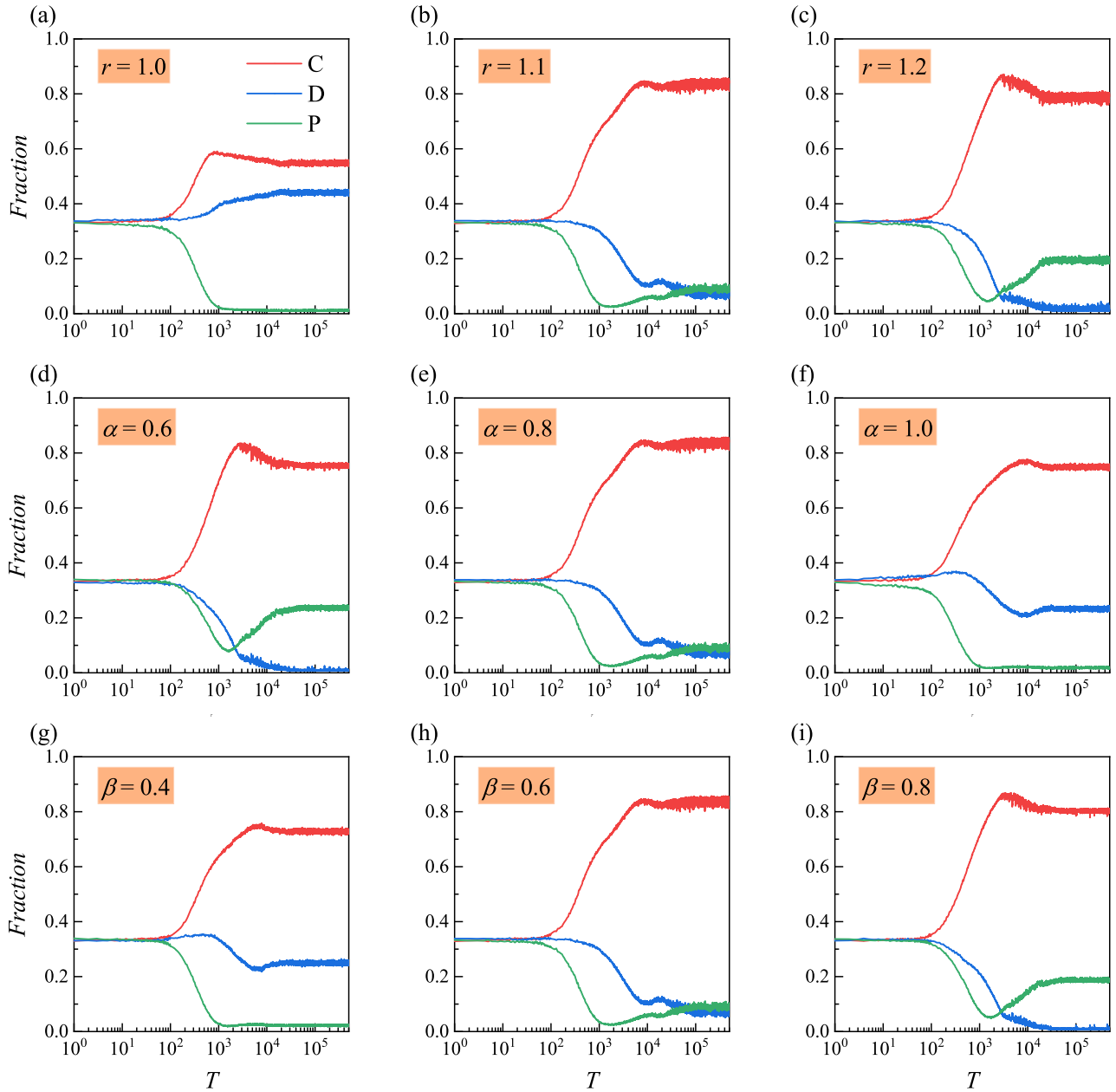


Fig. 3. Time courses of the fraction of three strategies for several values of r ($\alpha = 0.8, \beta = 0.6$), α ($r = 1.1, \beta = 0.6$), and β ($r = 1.1, \alpha = 0.8$). Other parameters are $g = 5$, $L = L_c$, $\langle k \rangle = k_c$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

as α increases. These observations suggest that increasing α reduces the benefits for punishers so that cooperators need a large synergy factor to resist the invasion of defectors, while increasing β reduces the benefits for defectors, making it easier for cooperators to become the dominant strategy in the population. In Fig. 2(b–c), for various values of α and β at $r = 1.1$, there exists an optimal value of α (β) that achieves the highest cooperation level. Furthermore, this optimal α (β) value increases with an increase in β (α). This suggests that maintaining a balanced competition between punishers and defectors can enhance cooperation. The reason is that increasing α diminishes the benefits of punishers, yet excessive costs can result in the elimination of punishment, exacerbating the influx of defectors into cooperative groups. Regarding the β scenario, increasing fines for defectors can foster cooperation. However, when r is high, punishers also see increased benefits. In a situation with a mild dilemma, a moderate β is optimal

for promoting cooperation. Conversely, if the dilemma is intense, higher fines must be imposed on defectors to sustain a high cooperation level.

In order to explain the optimal phenomena caused by the synergy factor r , cost α , and fine β , we first depict the evolution time courses of the three strategies for different values of r in Fig. 3(a–c). When $r = 1.0$, punishers are unable to access sufficient allocated payoffs to sustain their punitive actions, leading to their gradual disappearance and exacerbating the cooperation dilemma. When the system reaches the steady state, the fraction of defectors stabilizes at an elevated level, resulting in suppressed cooperation level within the population. As r increases to 1.1, the evolution of punishment strategy has the enduring period (END) and the expand period (EXP) (Perc et al., 2008; Szolnoki & Perc, 2008). Although the fraction of defectors and punishers decreases during the END phase, the number of punishers increases during the EXP phase, as

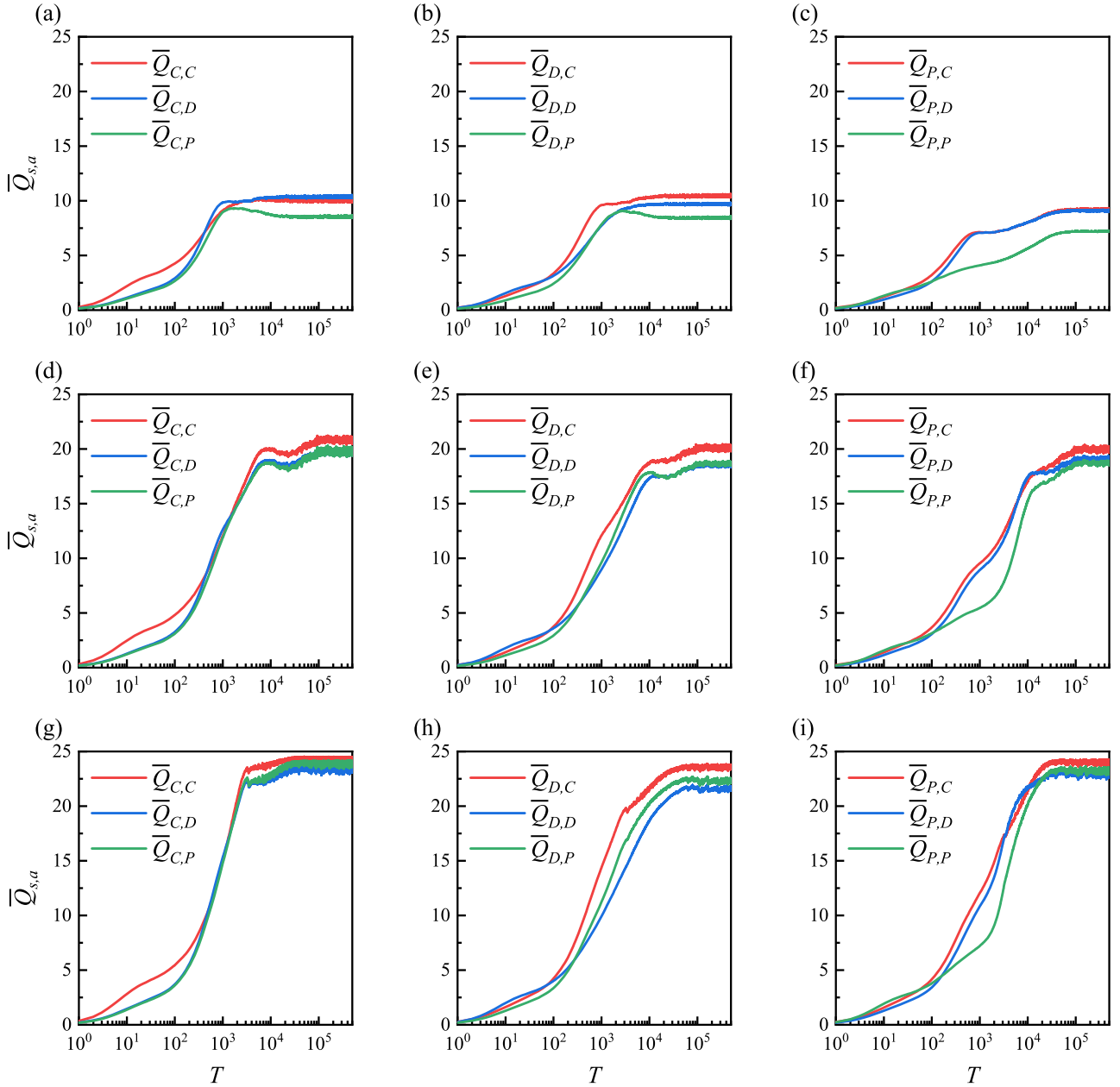


Fig. 4. Time courses of the average Q-value of different state-action pairs for several values of r . The rows from top to bottom represent the results of $r = 1.0$, $r = 1.1$ and $r = 1.2$, respectively. Other parameters are $\alpha = 0.8$, $\beta = 0.6$, $g = 5$, $L = L_c$, $\langle k \rangle = k_c$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

they incur no extra cost when the number of defectors is low across the whole population. When the system reaches the steady state, both defectors and punishers maintain minimal presence, enabling cooperators to reach their maximal population share. Consistent with the findings of Wang and Szolnoki (2023), we demonstrate that punishers can persist in the population even under relatively weak dilemma strength, despite punishment not being the immediately optimal strategy for individual payoff maximization. This strategy inertia enables punishers to continuously sanction defectors, ultimately driving the cooperation level to its maximum potential. However, when $r = 1.5$, the proportion of punishers remains constant throughout the entire evolutionary process, and the rivalry between cooperators and punishers intensifies. When the system reaches the steady state, the fraction of punishers stabilizes at an elevated level, resulting in suppressed cooperation level within the population. Furthermore, we have plotted the time evolution of the fractions for the three strategies across various α values to elucidate the rationale

behind the optimal cost. As depicted in Figs. 3(d–f), an increase in α results in a decrease of the punisher frequency, suggesting that punishers find it challenging to form clusters to gather sufficient benefits, thereby allowing defectors to persist within the population. In this context, the system's coexistence is marked by a combination of cooperators and defectors. Consequently, an overly high value of α facilitates the spread of defection, compelling cooperators to confront the encroachment of defectors resulting from the absence of sufficient punishment. At the steady state, the fraction of punishers monotonically decreases as α increases while defectors progressively dominate, resulting in a non-monotonic trend for cooperators that initially rises before declining. Furthermore, to elucidate the impact of β on optimal cooperation, Figs. 3(g–i) illustrate the temporal dynamics of the proportions of three strategies across various β values. These results indicate that increased punishment for defectors diminishes the prevalence of defection. As the system reaches equilibrium, with higher β values, the proportion of punishers ceases

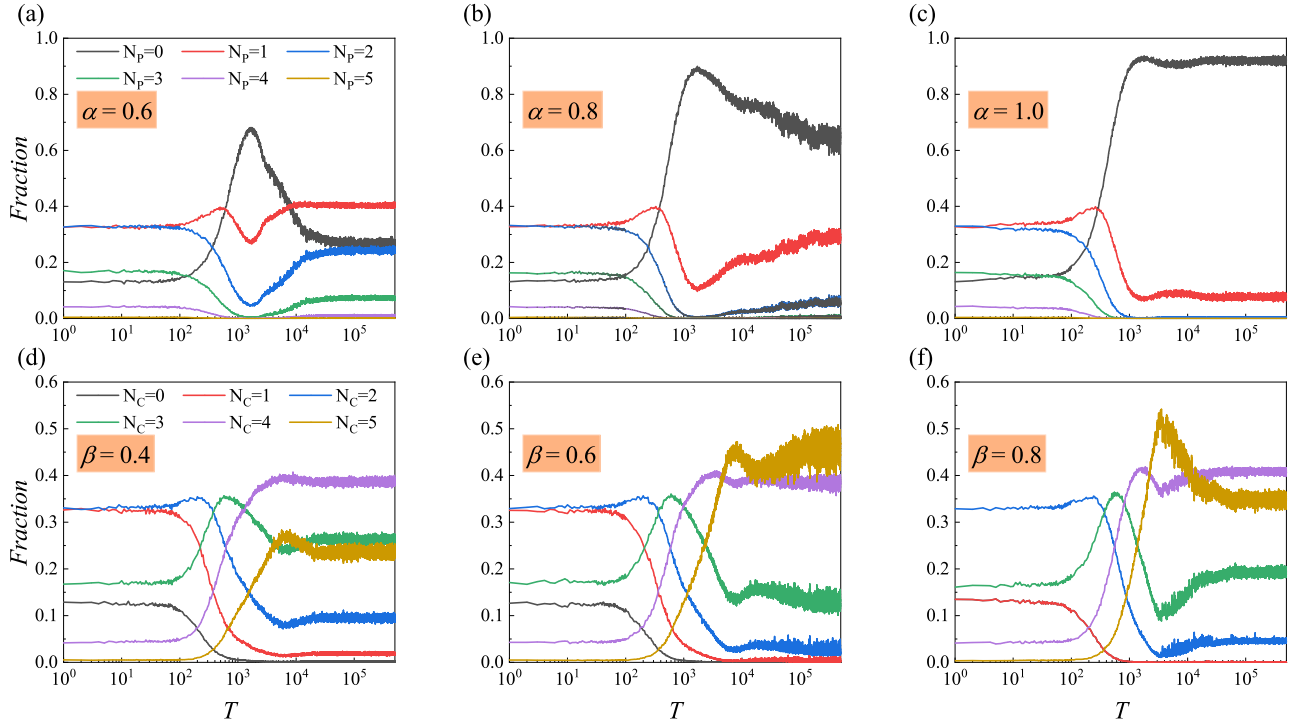


Fig. 5. Time courses of the punitive groups and cooperative groups for several values of α and β . Other parameters are $r = 1.1$, $g = 5$, $L = L_c$, $\langle k \rangle = k_c$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

its decline, ensuring a critical mass of punishers capable of curbing the free-riding behaviors of defectors (first-order free-riding behaviors) and cooperators (second-order free-riding behaviors). Nevertheless, excessive benefits for punishers can be detrimental to the propagation of cooperation, leading to a lower cooperation level at $\beta = 0.8$ compared to $\beta = 0.6$. At the steady state, an increase in β triggers a characteristic shift where defector prevalence undergoes sustained decline while punisher populations demonstrate consistent growth, ultimately generating the same non-monotonic cooperator dynamics.

Subsequently, we present the evolutionary trajectory of the average Q-value (i.e., $\bar{Q}_{s,a} = \sum_i^N Q_{s,a}^i / N$) across various state-action combinations to elucidate the dynamics of strategy transitions. As depicted in Fig. 4(a–c), during the END phase, agents predominantly opt for actions that align with their current state. In the EXP phase, while some cooperators become defectors, a greater number of defectors revert to cooperation, and the likelihood of punishers transitioning to cooperators becomes nearly equivalent by the end. At the steady state, the following average Q-value relationships are observed: $\bar{Q}_{C,D} > \bar{Q}_{C,C}$ ($\bar{Q}_{C,P}$), $\bar{Q}_{D,C} > \bar{Q}_{D,D}$ ($\bar{Q}_{D,P}$), and $\bar{Q}_{P,C} \approx \bar{Q}_{P,D}$. This suggests that agents are more inclined to choose defection when in a cooperation state, prefer cooperation when in a defection state, and tend to select either cooperation or defection when in a punishment state, which results in a higher frequency of cooperators relative to defectors, with the latter's numbers remaining at a moderate level. Due to the persistent presence of defectors, the cooperation level remains consistently low. Conversely, for the $r = 1.1$ scenario depicted in Fig. 4(d–f), the Q-value inequalities ($\bar{Q}_{C,C} > \bar{Q}_{C,D}$ ($\bar{Q}_{C,P}$), $\bar{Q}_{D,C} > \bar{Q}_{D,D}$ ($\bar{Q}_{D,P}$), and $\bar{Q}_{P,C} > \bar{Q}_{P,D}$ ($\bar{Q}_{P,P}$)) demonstrate that agents exhibit a consistent preference for cooperative behavior regardless of their current state. The majority of defectors and punishers are inclined towards becoming cooperators, while cooperators persist with their strategy, facilitating the formation of cooperative clusters. In this scenario, the frequencies of defectors and punishers remain relatively low, contributing to a higher overall cooperation level within the system. As r increases significantly, the propensity of cooperators to switch between being cooperators or punishers becomes comparable

($\bar{Q}_{C,C} \approx \bar{Q}_{C,P}$), as punishers can reap substantial benefits from a weaker dilemma intensity, and some punishers prefer to maintain their current strategy. This intensification of competition between cooperators and punishers, which results in a lower frequency of cooperators compared to the $r = 1.1$ case.

Moreover, to further elucidate the impact of parameters α and β on the optimal cooperation level, Fig. 5 illustrates the temporal evolution of the proportions of punitive and cooperative groups across varying α and β values, with N_C (N_P) denoting the count of cooperators (punishers) within a hyperlink. As depicted in Fig. 5(a–c), an increase in α is associated with a rise in the number of groups devoid of punishers, indicating that the presence of punishers diminishes with escalating costs. Furthermore, at $\alpha = 0.6$, a substantial presence of punishers is associated with reduced levels of cooperation, whereas the absence of punishers at $\alpha = 1.0$ facilitates the infiltration of defectors into cooperative clusters. Consequently, the optimal cooperation for α is realized when the ratio of punishers is modest yet adequate to deter defectors. Observing the outcomes presented in Fig. 5(d–f), the proportion of fully cooperative clusters is maximized at moderate β values. At $\beta = 0.4$, a minor penalty is insufficient to mitigate the competition between cooperators and defectors, whereas a substantial penalty empowers punishers to curtail the free-riding behavior of defectors and cooperators. Nonetheless, excessive benefits for punishers prove detrimental to the proliferation of cooperative behavior, leading to a lower stationary proportion of cooperators at $\beta = 0.8$ compared to $\beta = 0.6$.

Having examined the effect of the punishment mechanism on cooperation, we next investigate the influence of the self-regarding Q-learning algorithm on the evolution of cooperation. Fig. 6(a–c) illustrates that an increment in the learning rate η is detrimental to the promotion of cooperation, particularly when the discount factor γ is low. This finding suggests that a heightened focus on immediate game rewards tends to empower defectors, enabling them to exploit cooperators with greater efficacy. Conversely, when the algorithm considers a greater weight on expected future rewards, a moderate η facilitates the punisher's survival in the population, thereby intensifying

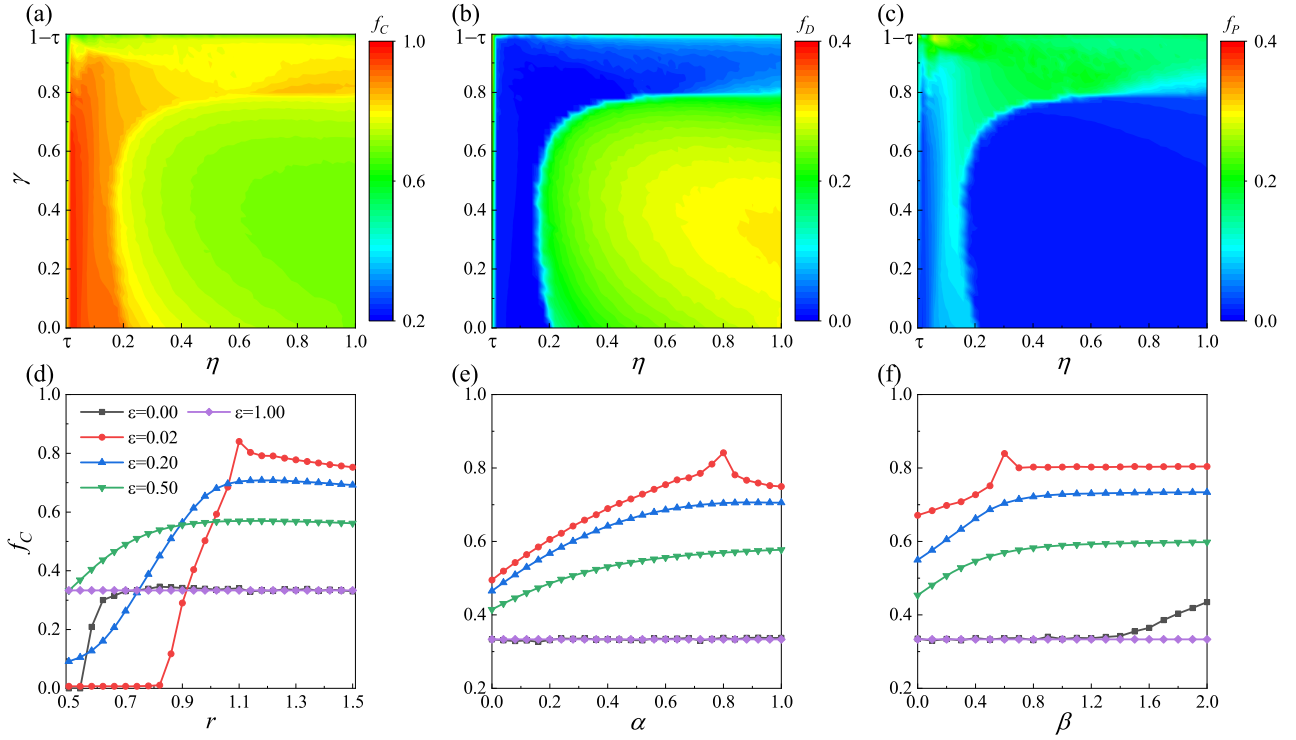


Fig. 6. (a–c) Phase diagrams of $\gamma - \eta$ for the evolution of three strategies. (d–f) Frequency of cooperators f_C as a function of r , α , and β for several values of ϵ . Other parameters are $g = 5$, $L = L_c$, $\langle k \rangle = k_c$, $\tau = 0.0025$.

the competition between cooperators and punishers. Additionally, we study the effect of exploration rate ϵ on the evolution of cooperation, as depicted in Fig. 6(d–f). When $\epsilon = 0.0$, agents base their actions solely on the maximum Q-value within the current state, resulting in agents to persist with their current strategies. Consequently, the cooperation level remains akin to that observed during the initial phase. Nevertheless, excessively high values of β can substantially diminish the incentives for defectors, leading to a gradual rise in the cooperation level as β continues to increase. Moreover, when ϵ is small, a certain degree of randomness can enable the system to escape from the local optimal solution, thereby reflecting the optimal cooperation phenomenon of r , α , and β . Agents choose action more randomly with increasing ϵ . Thus, the evolution of cooperation is jeopardized as ϵ increases and the cooperation level is also similar with those of initial period when $\epsilon = 1.0$.

To examine the effects of hypergraph on the evolution of collective cooperation, we provide the frequency of cooperators f_C is depended on the order g and the number of hyperlinks L on the URHs. As shown in Fig. 7, we find that there exists an optimal value of g that leads to the highest cooperation level when order g increases. The increase in hypergraph order g leads to a reduction in the number of hyperlinks and consequently decreases the network's average degree. While existing literature suggests that densely connected networks generally hinder cooperation maintenance, excessively sparse networks similarly impede the propagation of cooperative behavior. This dual constraint implies the existence of an optimal g value that maximizes the system's cooperation level, representing a balance between network connectivity and cooperation sustainability. Excessively high hypergraph orders force cooperators to expend more effort across multiple groups, thereby intensifying competition between cooperators and defectors. Besides, the increase in hyperlink density L enables an agent to participate in more groups, which leads to a higher probability of defectors encountering punishers and consequently enhances the cooperation level.

Finally, we provided the simulation results and theoretical analysis of the frequency of cooperators f_C , defectors f_D , and punishers f_P as

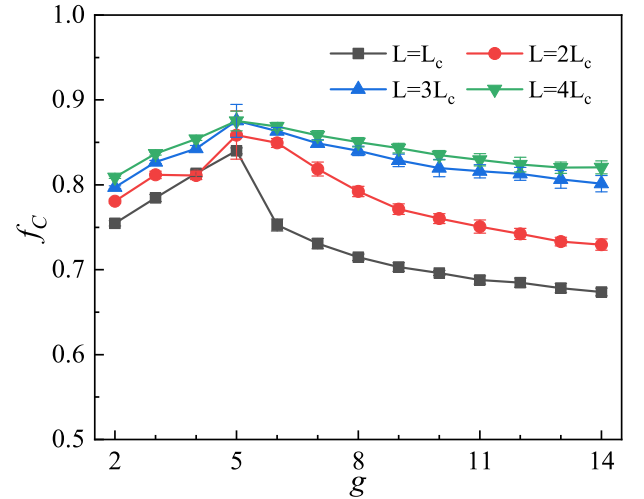


Fig. 7. Frequency of cooperators f_C as a function of order g varies for different number of hyperlinks L . Other parameters are $r = 1.1$, $\alpha = 0.8$, $\beta = 0.6$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

function of α and β . Here, we define the number of agents that update their strategy from a_x to a_y as $N_{a_x \rightarrow a_y}$ and the corresponding strategy transition probabilities as $W_{a_x \rightarrow a_y}$, which can be calculated as

$$W_{a_x \rightarrow a_y} = \frac{1}{N_{a_x}} N_{a_x \rightarrow a_y}, \quad (5)$$

where N_{a_x} is the number of agents who adopt strategy a_x before updating their actions. After that, we use the mean-field method to analyze the dynamics of the frequencies of cooperators (f_C), defectors (f_D) and punishers (f_P), which can be approximated as (Guo et al., 2022; Wang et al., 2023a)

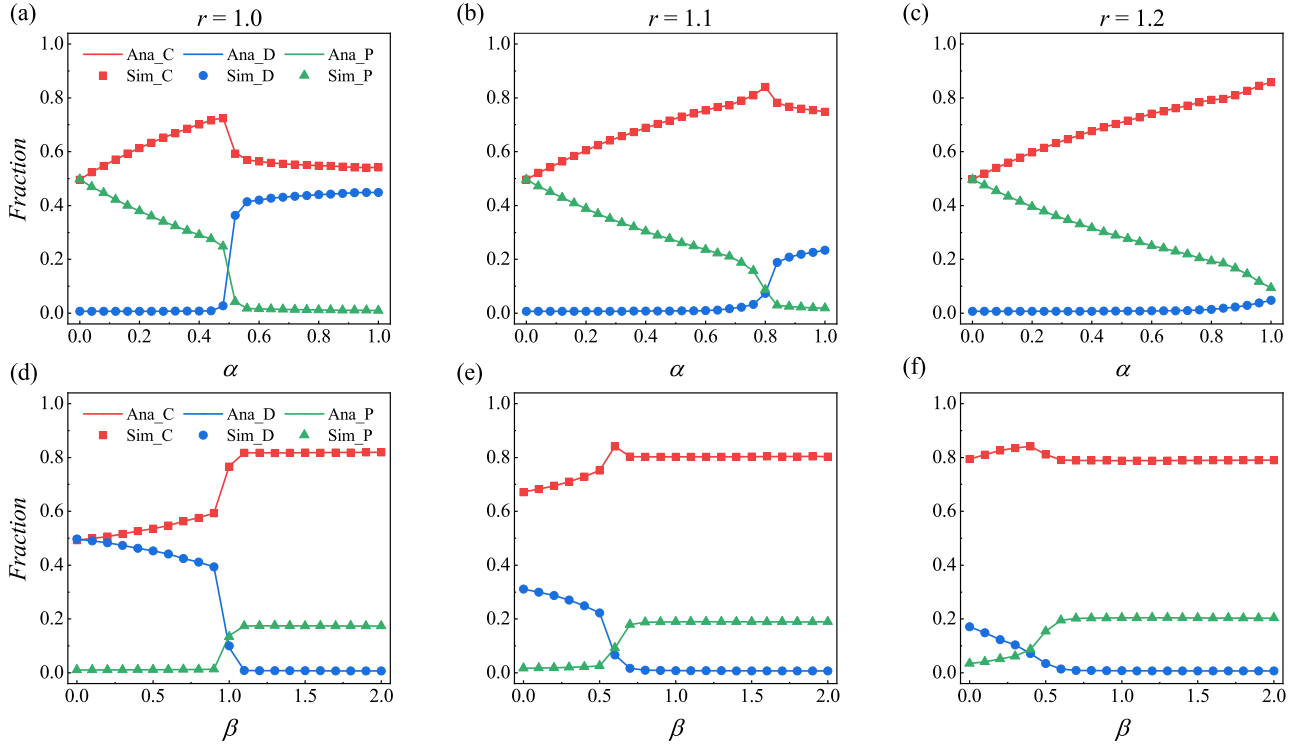


Fig. 8. Frequency of cooperators f_C , defectors f_D , and punishers f_P as a function of (a–c) α and (d–f) β for different values of r on the simulation results (dot) and theoretical analysis (line). Other parameters are $g = 5$, $L = L_c$, $\langle k \rangle = k_c$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

$$\begin{aligned}\dot{f}_C &= f_D W_{D \rightarrow C} + f_P W_{P \rightarrow C} - f_C (W_{C \rightarrow D} + W_{C \rightarrow P}), \\ \dot{f}_D &= f_C W_{C \rightarrow D} + f_P W_{P \rightarrow D} - f_D (W_{D \rightarrow C} + W_{D \rightarrow P}), \\ \dot{f}_P &= f_C W_{C \rightarrow P} + f_D W_{D \rightarrow P} - f_P (W_{P \rightarrow C} + W_{P \rightarrow D}).\end{aligned}\quad (6)$$

When the system reaches a steady state, all strategies have an update rate of 0, i.e., $\dot{f}_C = \dot{f}_D = \dot{f}_P = 0$. Then the specific f_C , f_D and f_P can be deduced as

$$\begin{aligned}\mu &= \frac{W_{C \rightarrow D} W_{P \rightarrow C} + W_{C \rightarrow D} W_{P \rightarrow D} + W_{C \rightarrow P} W_{P \rightarrow D}}{W_{D \rightarrow C} W_{P \rightarrow C} + W_{D \rightarrow C} W_{P \rightarrow D} + W_{D \rightarrow P} W_{P \rightarrow C}}, \\ f_C &= \frac{W_{P \rightarrow C} + W_{P \rightarrow D}}{(1 + \mu) * (W_{P \rightarrow C} + W_{P \rightarrow D}) + W_{C \rightarrow P} + \mu * W_{D \rightarrow P}}, \\ f_D &= f_C * \mu, \\ f_P &= f_C \frac{W_{C \rightarrow P} + \mu * W_{D \rightarrow P}}{W_{P \rightarrow C} + W_{P \rightarrow D}}.\end{aligned}\quad (7)$$

We depict the results of numerical simulation and theoretical analysis in Fig. 8, where we can observe that the theoretical results are basically consistent with the numerical results. Besides, it is worth noting that the mean-field approximation primarily computes the theoretical average payoffs of cooperators and defectors in well-mixed populations and derives the dynamical relationship between cooperation levels and system parameters through replicator dynamics. This conventional approach inherently neglects the influence of network structure in its formulation. Consequently, the incorporation of more realistic and complex network structures into the mean-field approximation framework has remained a persistent challenge in the field, as the existing methodology struggles to account for these structural complexities while maintaining analytical tractability.

To better understand the impact of the self-regarding Q-learning algorithm on the evolution of cooperation, we discuss how self-regarding Q-learning algorithm affect cooperation dynamics compared to Fermi update rule. As shown in Fig. 9, the Fermi update rule does not yield an identifiable synergy factor r that would optimize cooperation levels. Under conditions where $r > 1$, cooperators are able to accumu-

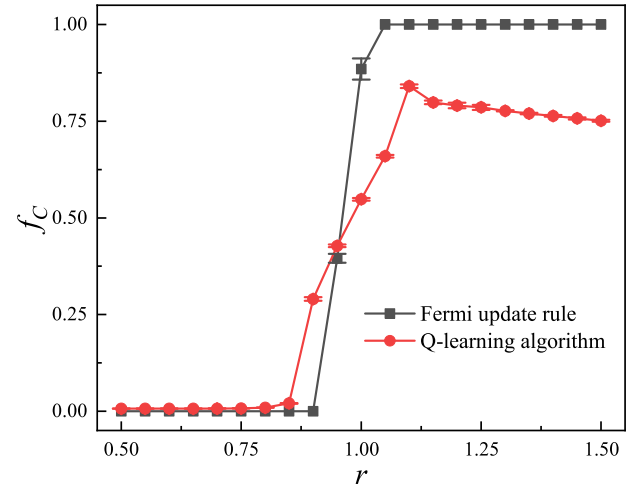


Fig. 9. Frequency of cooperators f_C as a function of r under self-regarding Q-learning algorithm and Fermi update rule. Other parameters are $r = 1.1$, $\alpha = 0.8$, $\beta = 0.6$, $\kappa = 0.1$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

late sufficient payoffs to maintain cooperative behavior, while the neighbor-imitation mechanism promotes the widespread dissemination of cooperation, eventually resulting in universal adoption of cooperative strategies throughout the population. For the self-regarding Q-learning algorithm results, even when $r > 1$, the inherent stochasticity of the ϵ -greedy exploration method prevents convergence to local optima in agent behavioral choices, thereby allowing for the emergence of these optimal cooperation patterns. Consequently, we adopt the self-regarding Q-learning algorithm as our primary strategy update mechanism, enabling systematic investigation of these cooperative phenomena.

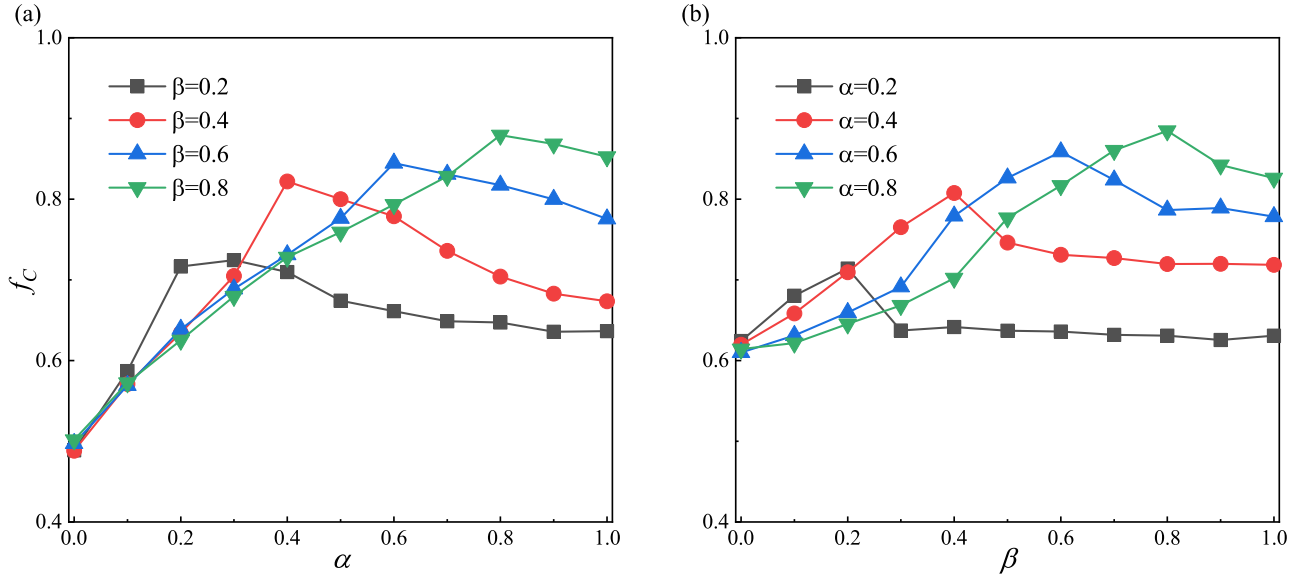


Fig. 10. Frequency of cooperators f_C as a function of α and β for several values of another parameter under empirical hypergraph data sets (Congress-bills Fowler, 2006). Other parameters are $g = 5$, $L = L_c$, $\langle k \rangle = k_c$, $r = 1.1$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

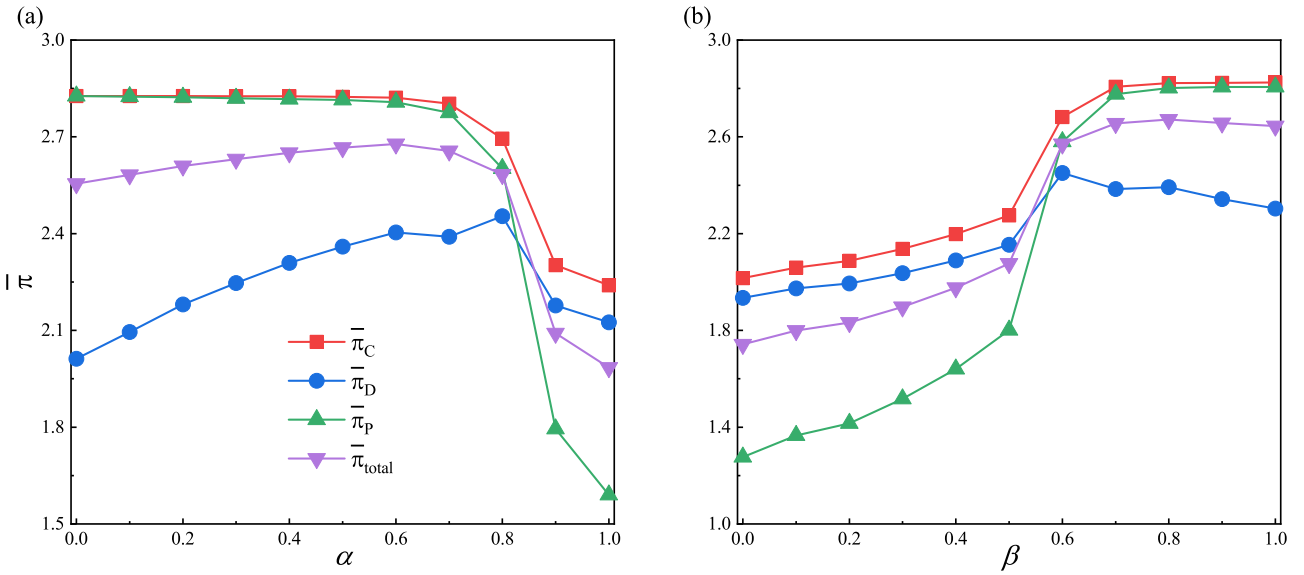


Fig. 11. Average payoffs of cooperators, defectors, punishers, and total as a function of α and β for several values of another parameter. Other parameters are $g = 5$, $L = L_c$, $\langle k \rangle = k_c$, $r = 1.1$, $\eta = 0.8$, $\gamma = 0.8$, and $\epsilon = 0.02$.

To validate the effectiveness of our model across diverse network structures, we constructed empirical networks based on both the original Flower data sets (Fowler, 2006) and the processed data sets from Iacopini et al. (2022), subsequently investigating the evolution of cooperative behavior under punishment mechanisms in these networks. As shown in Fig. 10, the results reveal that cooperation levels initially increase and then decrease with rising values of α and β , resulting in an optimal cooperation level. This phenomenon demonstrates that our model successfully captures the optimal cooperation patterns under varying combinations of punishment costs and fines, even in empirical network structures, thereby confirming the robustness of our findings across different network topologies.

We further examines the conditions under which punishment mechanisms enhance social welfare from the perspective of average pay-

offs. As illustrated in Fig. 11, increasing the punishment cost α leads to a gradual decline in punishers' average payoffs, ultimately causing their disappearance from the population. This dynamic subsequently elevates defectors' average payoffs and consequently increases their population proportion. The optimal cooperation phenomenon emerges when the differentials between the three strategies' average payoffs are minimized ($\alpha = 0.8$). Further increases in punishment costs beyond this point trigger a sharp decline in cooperators' average payoffs, leading to reduced cooperation levels. A similar pattern is observed for fine β , where optimal cooperation occurs when payoff differences are minimized ($\beta = 0.6$). These findings suggest that social resource allocators should aim to minimize inter-group payoff disparities when determining punishment costs and fines, as this approach effectively enhances overall social welfare.

4. Conclusion

In this study, we propose a spatial public goods game model on uniform random hypergraphs, incorporating punishment as a mechanism. In our model, punishers contribute to the common pool, similar to cooperators, and incur a cost to penalize defectors, who are subsequently fined. Agents are assumed to update their strategies using a self-regarding Q-learning algorithm to maximize individual benefits. Through Monte Carlo simulations, we first explore the evolutionary dynamics of various strategies, examining how they are influenced by parameter pairs drawn from the synergy factor, cost, and fine. Subsequently, we seek to explain the emergence of optimal cooperation phenomena and investigate the impact of endogenous parameters within the self-regarding Q-learning algorithm on the evolution of cooperation. Ultimately, we present a comparison between theoretical analysis and numerical simulations to establish the effectiveness of our model. Our findings suggest that a moderate synergy factor can optimally facilitate communication among cooperative agents. Furthermore, we identify cost-fine pairs that can achieve the highest proportion of cooperators in environments with a large synergy factor. Additionally, we observe that increased decision-making randomness negatively impacts cooperation. These findings provide valuable guidance for the allocation and utilization of public resources under punishment mechanisms. For instance, in judicial decision-making contexts, sentencing duration may be conceptualized as an imposed penalty, while the societal impact of the verdict constitutes the associated punitive cost. Our analysis suggests that disproportionately severe sentences may impose unsustainable financial burdens on defendants, potentially triggering extreme responses such as self-harm or escape attempts, while simultaneously generating detrimental societal consequences. Conversely, excessively lenient sentences reduce the opportunity cost of criminal behavior, potentially fostering social instability and producing similarly adverse societal effects. This framework consequently points to the necessity of identifying an intermediate sentencing range that maintains proportionality between penalties and defendants' capacity while optimizing positive social outcomes. The theoretical insights derived from our study may provide valuable guidance for achieving this delicate balance in judicial practice. In future studies, we plan to extend this model to more complex networks (e.g., power random hypergraphs [Alvarez-Rodriguez et al., 2021](#)) to examine how node centrality affect strategic interactions in higher-order network ([Cardoso et al., 2020](#)), and explore the effects of asymmetric interactions in the context of spatial public goods games ([Su et al., 2021](#)). Besides, we also can further examine the influence of personal and institutional characteristics on the evolution of cooperation ([Benchimol et al., 2022](#)) or investigate the different effect between direct punishment and sampling punishment ([Xiao et al., 2023](#)).

CRedit authorship contribution statement

Kuan Zou: Formal analysis, Investigation, Data curation, Visualization, Writing – original draft; **Changwei Huang:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – review & editing, Funding acquisition.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Allen, B., Lippner, G., Chen, Y.-T., Fotouhi, B., Momeni, N., Yau, S.-T., & Nowak, M. A. (2017). Evolutionary dynamics on any population structure. *Nature*, 544(7649), 227–230.
- Alvarez-Rodriguez, U., Battiston, F., Arruda, G. F. D., Moreno, Y., Perc, M., & Latora, V. (2021). Evolutionary dynamics of higher-order interactions in social networks. *Nature Human Behaviour*, 5, 586–595.
- Axelrod, R., & Hamilton, W. D. (1981). The evolution of cooperation. *Science*, 211(4489), 1390–1396.
- Battiston, F., Amico, E., Barrat, A., Bianconi, G., Arruda, G. F. D., Franceschiello, B., Iacopini, I., Kéfi, S., Latora, V., Moreno, Y., Murray, M. M., Peixoto, T. P., Vaccarino, F., & Petri, G. (2021). The physics of higher-order interactions in complex systems. *Nature Physics*, 10, 1093–1098.
- Battiston, F., Cencetti, G., Iacopini, I., Latora, V., Lucas, M., Patania, A., Young, J.-G., & Petri, G. (2020). Networks beyond pairwise interactions: structure and dynamics. *Physics Reports*, 874, 1–92.
- Benchimol, J., El-Shagi, M., & Saadon, Y. (2022). Do expert experience and characteristics affect inflation forecasts? *Journal of Economic Behavior & Organization*, 201, 205–226.
- Benchimol, J., & Palumbo, L. (2024). Sanctions and russian online prices. *Journal of Economic Behavior & Organization*, 225, 483–521.
- Burgio, G., Matamalas, J. T., Gómez, S., & Arenas, A. (2020). Evolution of cooperation in the presence of higher-order interactions: From networks to hypergraphs. *Entropy*, 22(7), Article 744.
- Cardoso, F. M., Gracia-Lázaro, C., Moisan, F., & Goyal, S. (2020). Effect of network topology and node centrality on trading. *Scientific Reports*, 10, Article 11113.
- Chen, X., Szolnoki, A., & Perc, M. (2014). Probabilistic sharing solves the problem of costly punishment. *New Journal of Physics*, 16, Article 83016.
- Civilini, A., Anbarci, N., & Latora, V. (2021). Evolutionary game model of group choice dilemmas on hypergraphs. *Physical Review Letters*, 127(26), Article 268301.
- Flores, L. S., Fernandes, H. C. M., Amaral, M. A., & Vainstein, M. H. (2021). Symbiotic behaviour in the public goods game with altruistic punishment. *Journal of Theoretical Biology*, 524, Article 110737.
- Fowler, J. H. (2006). Legislative cosponsorship networks in the US House and senate. *Social networks*, 28(4), 454–465.
- Fu, F., Hauert, C., Nowak, M. A., & Wang, L. (2008). Reputation-based partner choice promotes cooperation in social networks. *Physical Review E*, 78(2), Article 26117.
- Gianetto, D. A., & Heydari, B. (2016). Sparse cliques trump scale-free networks in coordination and competition. *Scientific Reports*, 6, Article 21870.
- Guo, H., Wang, Z., Song, Z., Yuan, Y., Deng, X., & Li, X. (2022). Effect of state transition triggered by reinforcement learning in evolutionary prisoner's dilemma game. *Neurocomputing*, 511, 187–197.
- Gürer, O., Irlenbusch, B., & Rockenbach, B. (2006). The competitive advantage of sanctioning institutions. *Science*, 312(5770), 108–111.
- Han, D., & Li, X. (2022). On evolutionary vaccination game in activity-driven networks. *IEEE Transactions on Computational Social Systems*, 10(1), 142–152.
- Hauert, C., & Doebeli, M. (2004). Spatial structure often inhibits the evolution of cooperation in the snowdrift game. *Nature*, 428(6983), 643–646.
- Helbing, D., Szolnoki, A., Perc, M., & Szabó, G. (2010). Punish, but not too hard: How costly punishment spreads in the spatial public goods game. *New Journal of Physics*, 12(8), 83005.
- Henrich, J., McElreath, R., Barr, A., Ensminger, J., Barrett, C., Bolyanatz, A., Cardenas, J. C., & Gurven, M. (2006). Costly punishment across human societies. *Science*, 312(5781), 1767–1770.
- Horita, Y., Takezawa, M., Inukai, K., Kita, T., & Masuda, N. (2017). Reinforcement learning accounts for moody conditional cooperation behavior: experimental results. *Scientific Reports*, 7(1), 39275.
- Hua, S., & Liu, L. (2024). Coevolutionary dynamics of population and institutional rewards in public goods games. *Expert Systems with Applications*, 237, Article 121579.
- Iacopini, I., Petri, G., Baronchelli, A., & Barrat, A. (2022). Group interactions modulate critical mass dynamics in social convention. *Communications Physics*, 5, 64.
- Jia, D., Guo, H., Song, Z., Shi, L., Deng, X., Perc, M., & Wang, Z. (2021). Local and global stimuli in reinforcement learning. *New Journal of Physics*, 23(8), Article 083020.
- Kumar, A., Chowdhary, S., Capraro, V., & Perc, M. (2021). Evolution of honesty in higher-order social networks. *Physical Review E*, 104(5), Article 54308.
- Liu, L., Chen, X., & Szolnoki, A. (2017). Competitions between prosocial exclusions and punishments in finite populations. *Scientific Reports*, 7(1), Article 46634.

- Liu, L., Wang, S., Chen, X., & Perc, M. (2018). Evolutionary dynamics in the public goods games with switching between punishment and exclusion. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 28(10), Article 103105.
- Ma, X., Quan, J., & Wang, X. (2023). Effect of reciprocity mechanisms on evolutionary dynamics in feedback-evolving games. *Nonlinear Dynamics*, 112, 709–729.
- Macy, M. W., & Flache, A. (2002). Learning dynamics in social dilemmas. *Proceedings of the National Academy of Sciences*, 99, 7229–7236.
- Majhi, S., Perc, M., & Ghosh, D. (2022). Dynamics on higher-order networks: A review. *Journal of the Royal Society Interface*, 19(188), Article 20220043.
- Nakamaru, M., & Yokoyama, A. (2020). The effect of ostracism and optional participation on the evolution of cooperation in the voluntary public goods game. *PLOS ONE*, 15(4), Article e0230973.
- Nowak, M. A., & May, R. M. (1992). Evolutionary games and spatial chaos. *Nature*, 359(6398), 826–829.
- Pan, J., Zhang, L., Han, W., & Huang, C. (2023). Heterogeneous investment promotes cooperation in spatial public goods game on hypergraphs. *Physica A*, 609, Article 128400.
- Pendharkar, P. C. (2012). Game theoretical applications for multi-agent systems. *Expert Systems with Applications*, 39, 273–279.
- Perc, M., Jordan, J. J., Rand, D. G., Wang, Z., Boccaletti, S., & Szolnoki, A. (2017). Statistical physics of human cooperation. *Physics Reports*, 687(8), 1–51.
- Perc, M., Szolnoki, A., & Szabó, G. (2008). Restricted connections among distinguished players support cooperation. *Physical Review E*, 78(6), Article 066101.
- Quan, J., Chen, X., Yang, W., & Wang, X. (2023). Cooperation dynamics in spatial public goods games with graded punishment mechanism. *Nonlinear Dynamics*, 111, 8837–8851.
- Rand, D. G., & Nowak, M. A. (2013). Human cooperation. *Trends in Cognitive Sciences*, 17(8), 413–425.
- Santos, F. C., Santos, M. D., & Pacheco, J. M. (2008). Social diversity promotes the emergence of cooperation in public goods games. *Nature*, 454(7201), 213–216.
- Santos, M. D., Pinheiro, F. L., Santos, F. C., & Pacheco, J. M. (2012). Dynamics of n-person snowdrift games in structured populations. *Journal of Theoretical Biology*, 315(2012), 81–86.
- Shi, Y., & Rong, Z. (2022). Analysis of q-learning like algorithms through evolutionary game dynamics. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 69(5), 2463–2467.
- Sigmund, K. (2007). Punish or perish? Retaliation and collaboration among humans. *Trends in Ecology & Evolution*, 22(11), 593–600.
- Song, Z., Guo, H., Jia, D., Perc, M., Li, X., & Wang, Z. (2022). Reinforcement learning facilitates an optimal interaction intensity for cooperation. *Neurocomputing*, 513, 104–113.
- Su, Q., Allen, B., & Plotkin, J. B. (2021). Evolution of cooperation with asymmetric social interactions. *Proceedings of the National Academy of Sciences*, 119(1), Article e2113468118.
- Szabó, G., & Fath, G. (2007). Evolutionary games on graphs. *Physics Reports*, 446, 97–216.
- Szolnoki, A., & Perc, M. (2008). Promoting cooperation in social dilemmas via simple coevolutionary rules. *The European Physical Journal B*, 67, 337–344.
- Szolnoki, A., & Perc, M. (2010). Reward and cooperation in the spatial public goods game. *EPL*, 92(3), Article 39003.
- Szolnoki, A., Szabó, G., & Perc, M. (2011). Phase diagrams for the spatial public goods game with pool punishment. *Physical Review E*, 83(3), Article 036101.
- Wang, C., & Szolnoki, A. (2023). Inertia in spatial public goods games under weak selection. *Applied Mathematics and Computation*, 449, Article 127941.
- Wang, L., Fang, L., Zhang, L., Zou, R., & Wang, Z. (2023a). Synergistic effects of adaptive reward and reinforcement learning rules on cooperation. *New Journal of Physics*, 25, Article 073008.
- Wang, Z., Jusup, M., Guo, H., Shi, L., Gecek, S., Anand, M., Perc, M., Bauchk, C. T., Kurths, J., Boccaletti, S., & Schellnhuber, H. J. (2020). Communicating sentiment and outlook reverses inaction against collective risks. *Proceedings of the National Academy of Sciences*, 117(30), 17650–17655.
- Wang, Z., Jusup, M., Shi, L., Lee, J.-H., Iwasa, Y., & Boccaletti, S. (2018). Exploiting a cognitive bias promotes cooperation in social dilemma experiments. *Nature Communications*, 9, Article 2954.
- Wang, Z., Jusup, M., Wang, R.-W., Shi, L., Iwasa, Y., Moreno, Y., & Kurths, J. (2017). Onymity promotes cooperation in social dilemma experiments. *Science Advances*, 3(3), Article e1601444.
- Wang, Z., Li, R., Jin, X., & Ding, H. (2022). Emergence of social norms in metanorms game with high-order interaction topology. *IEEE Transactions on Computational Social Systems*, 10(3), 1057–1072.
- Wang, Z., Song, Z., Shen, C., & Hu, S. (2023b). Emergence of punishment in social dilemma with environmental feedback. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37, 11708–11716.
- Wang, Z., Szolnoki, A., & Perc, M. (2014). Rewarding evolutionary fitness with links between populations promotes cooperation. *Journal of Theoretical Biology*, 349, 50–56.
- Watkins, C. J. C. H., & Dayan, P. (1992). Q-learning. *Machine Learning*, 8, 279–292.
- Xia, C., Wang, J., Perc, M., & Wang, Z. (2023). Reputation and reciprocity. *Physics of Life Reviews*, 46, 8–45.
- Xiao, J., Liu, L., Chen, X., & Szolnoki, A. (2023). Evolution of cooperation driven by sampling punishment. *Physics Letters A*, 475, Article 128879.
- Yang, H.-X., & Yang, J. (2019). Reputation-based investment strategy promotes cooperation in public goods games. *Physica A*, 523(1), 886–893.
- Zhang, L., & Huang, C. (2023). Preferential selection to promote cooperation on degree-degree correlation networks in spatial snowdrift games. *Applied Mathematics and Computation*, 454(1), 128113.
- Zou, K., Han, W., Zhang, L., & Huang, C. (2024). The spatial public goods game on hypergraphs with heterogeneous investment. *Applied Mathematics and Computation*, 466, Article 128450.
- Zou, K., & Huang, C. (2024). Incorporating reputation into reinforcement learning can promote cooperation on hypergraphs. *Chaos, Solitons & Fractals*, 186, Article 115203.